

SAHRDAYA COLLEGE OF ENGINEERING AND TECHNOLOGY

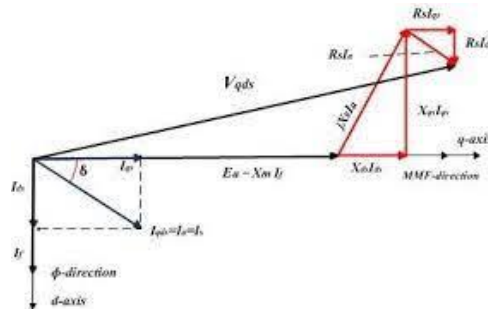
EET 307

Scheme

1. Single layer, double layer ,
open type, closed type ,
short pitched full pitched
2. Causes of Harmonics
Magnetic saturation
Linear load
Non uniform air gap etc

Elimination of harmonics
Distributed winding
Short chording
Skewing
Large airgap

3. X_d and X_q are present



4. Requirements:
Load sharing
Continuous supply
Maintenance etc

Conditions:

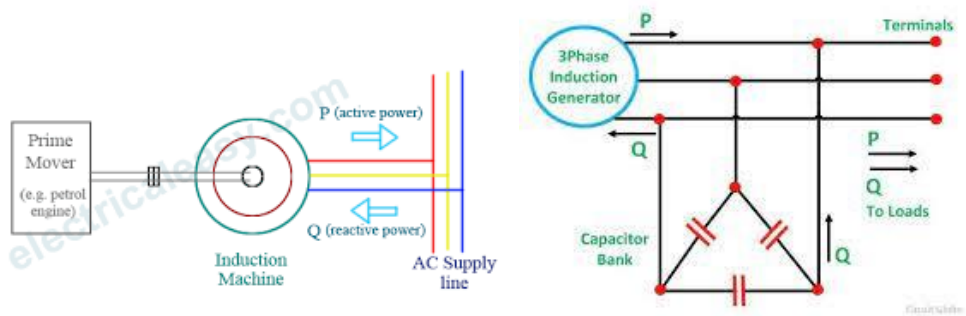
The voltage of the incoming alternator must be same as the bus-bar voltage.

The frequency of the incoming alternator must be same as the frequency of bus-bar voltage.

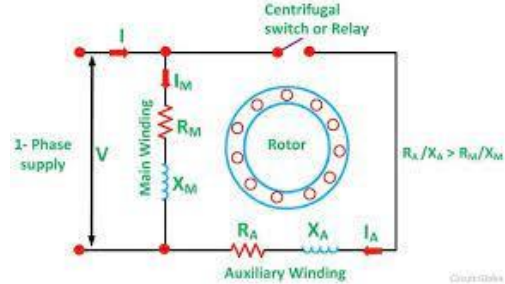
The phase of incoming alternator voltage must be identical to the phase angle of bus-bar voltage.

5. Damper winding start, pony motor start
6. $T = k_s \frac{E^2 R_2}{R_2^2 + X_2^2} s = 1$
7. star - delta starter diagram and explanation
8. V/F is abbreviated from voltage/frequency. V/F control is an induction motor control method which ensures the output voltage proportional with the frequency, so it maintains a constant motor flux, preventing weak magnetic and magnetic saturation phenomenon from happening.

9. Two modes explanation



10.



PART B

11. A)

easy insulation,
ease to current collection,
ventilation
Less sparks
less maintenance etc

B) $E_{ph} = 4.44 \cdot \phi \cdot T_{ph} \cdot K_p \cdot K_d$

$T_{ph} = 90$, $e_{mf \text{ phase}} = 1078$, $e_{mf \text{ line}} = 1868V$

12. A) $K_w = K_p \cdot K_d$

Derive each term using phasor diagrams

B) two types cylindrical and salient types with differences

13. OCC and SCC (4 marks) finding I_{f1} and I_{f2}

$I_{f1} = 32.5A$, $I_{f2} = 28A$,

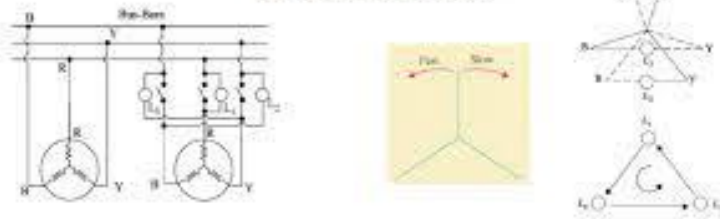
$F_r = 27.38A$

Phasor diagram

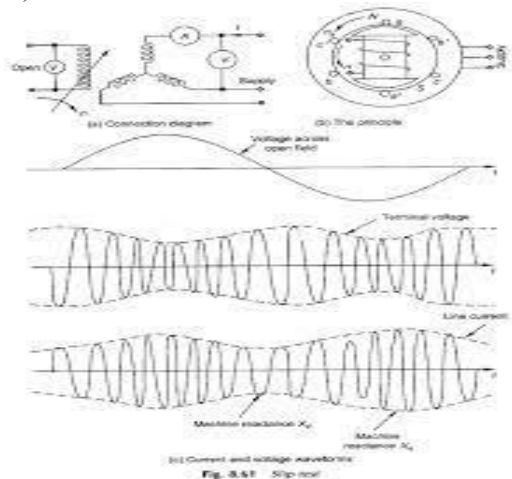
14. A)

2 Bright & 1 Dark Lamp method for Synchronization of Alternators

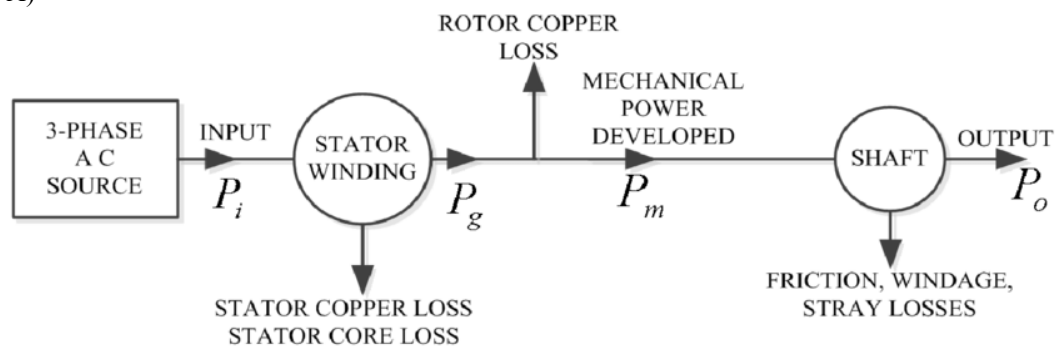
Prof. Jimit K. Shukla



B)aa

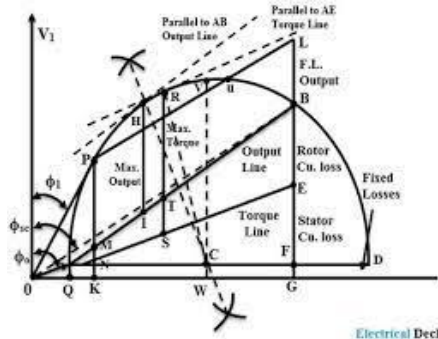


15. A) power supplied is 81.08Kw (2marks), armature current =146.3A,kw (2marks),so internal power developed =78.512kw (3 marks)
 b) power angle curve with δ vs power output
 16. A)



- B) Slip =.025, N_s =1000rpm, Power input to rotor=39kW, P_{mech} =38.025Kw, P_{out} =36.025Kw, η = 90.025,BHP=36.025kw/735=49(1 mark each

17.



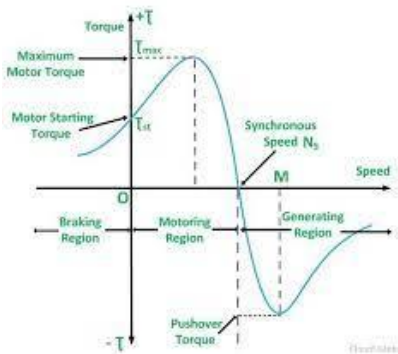
18.

- a) Plugging
- Dynamic Breaking
- Regenerative Breaking

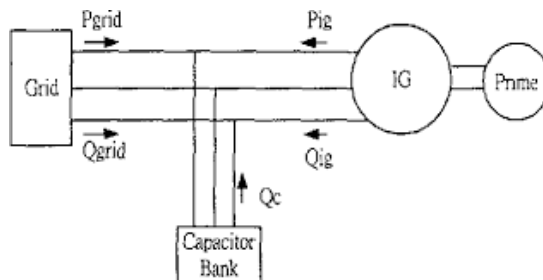
B) Resistance and reactance of outer and inner cages at starting and running condition for double cage induction motor has to be clearly explained with torque slip characteristics

19. A) Single phase supply is given means pulsating wave form only. It is necessary to create rotating magnetic field for rotation of rotor .

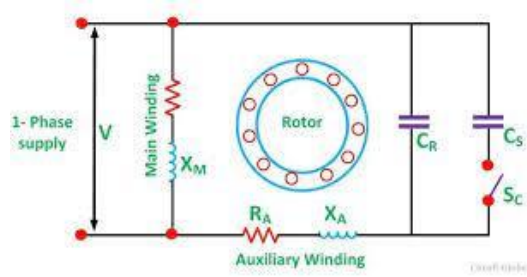
b)



20. A)



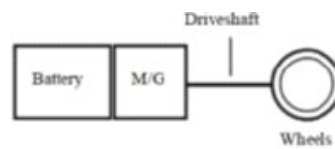
B)



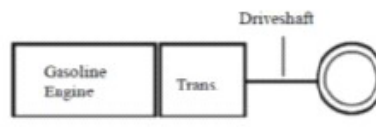
PART B

1. a) With explanation

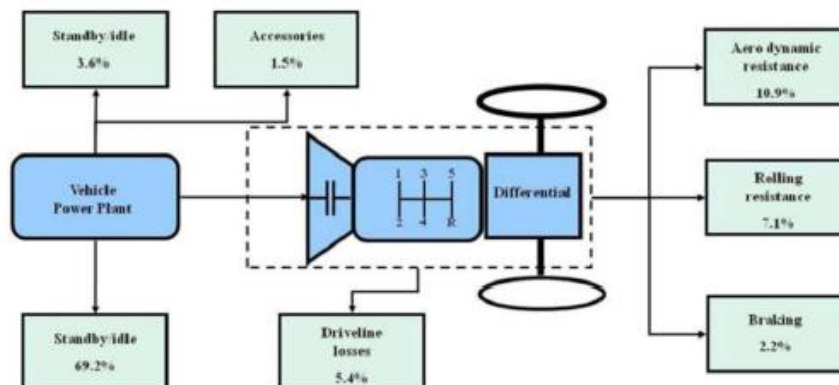
Pure Electrical



Pure Gasoline



b)



2.

a) With derivation

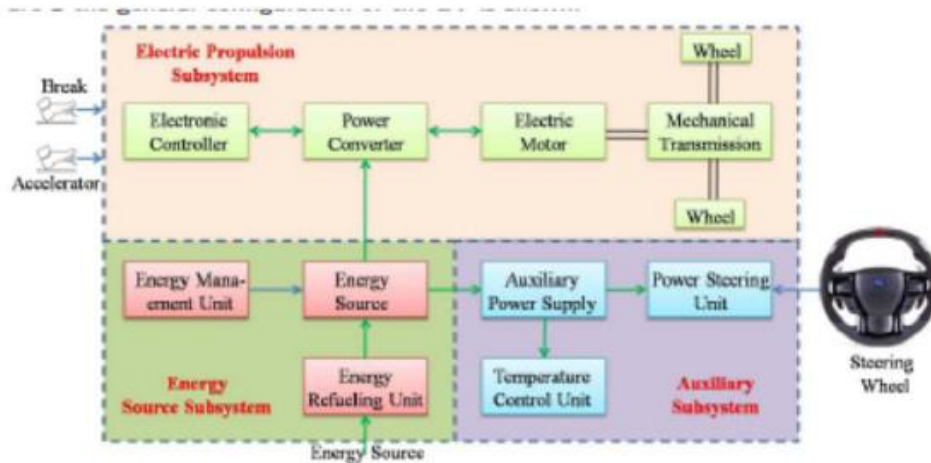
$$a = \frac{dV}{dt} = \frac{F_t - F_f - F_w}{\delta M}$$

$$= \frac{T_p i g i_o \eta_t / r_{dyn} - M g f_r - 1/2 \rho_a C_D A_f V^2}{\delta M} = \frac{g}{\delta} (d - f_r)$$

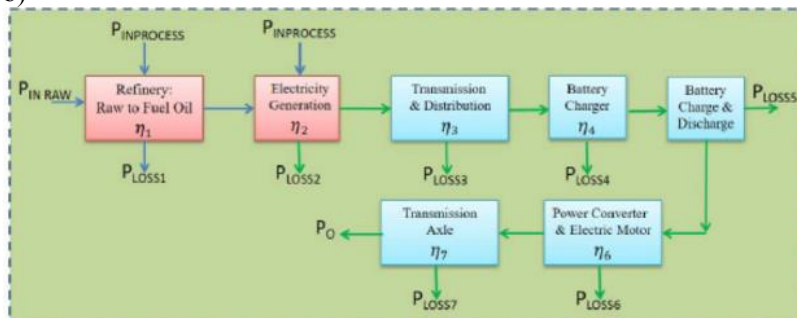
b)

- Optimal ICE operating point:
 - The optimal operating point on the torque-speed plane of the ICE can be based on maximization of fuel economy, the minimization of emissions or a compromise between fuel economy and emissions.
- Optimal ICE operating line:
 - In case the ICE needs to deliver different power demands, the corresponding optimal operating points constitute an optimal operating line.
- Safe battery voltage
 - The battery voltage may be significantly altered during discharging, generator charging or regenerative charging.
 - This battery voltage should not exceed the maximum voltage limit nor should it fall below the minimum voltage limit.

3. a)

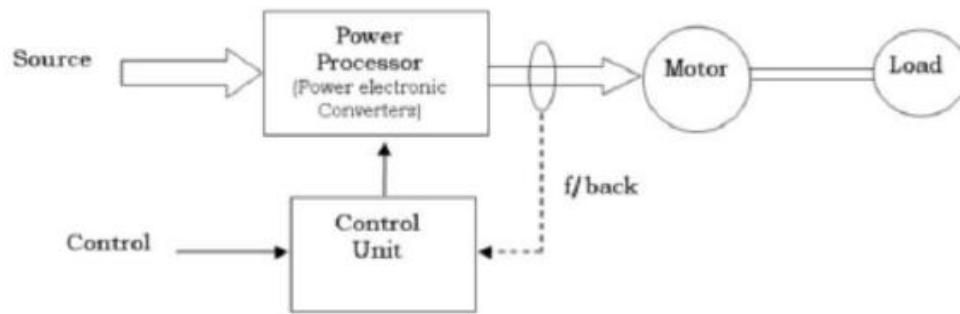


b)

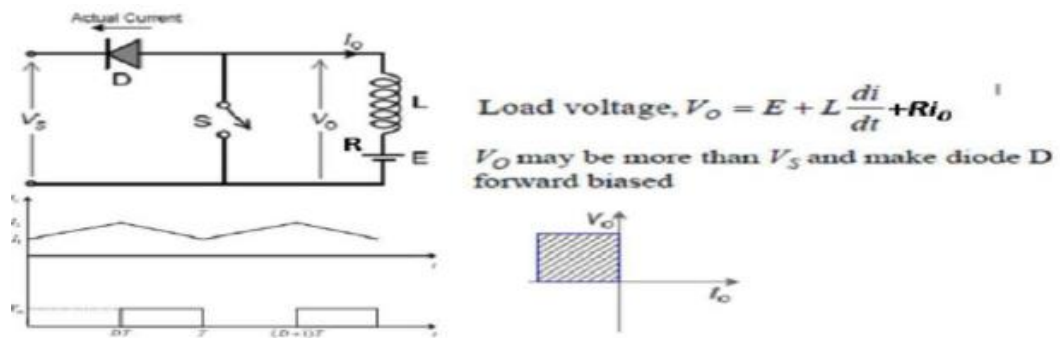


Part C

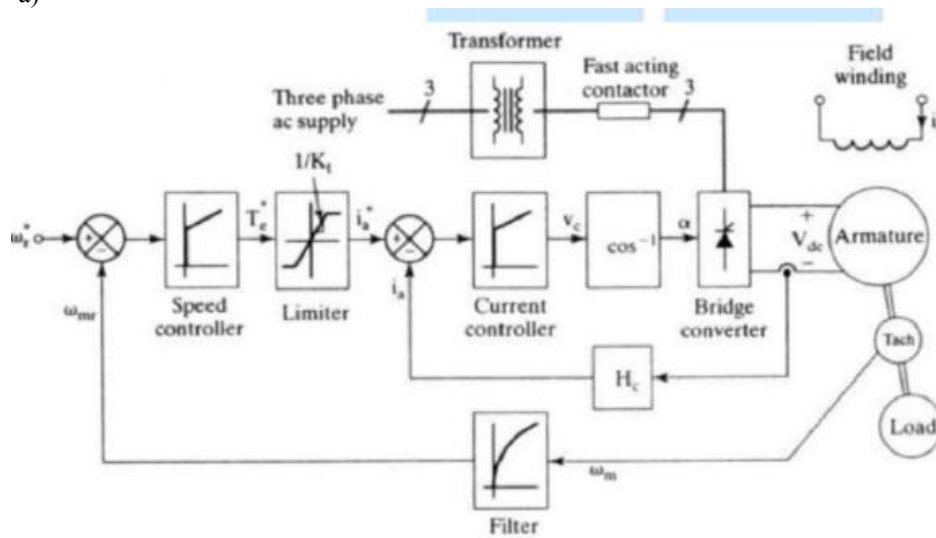
4. a)



b) write the equations also



5. a)



b) Explain – 4

6.

a) Fuel Cells: Basic principle – 2 marks

Advantages – Two points - 2 marks

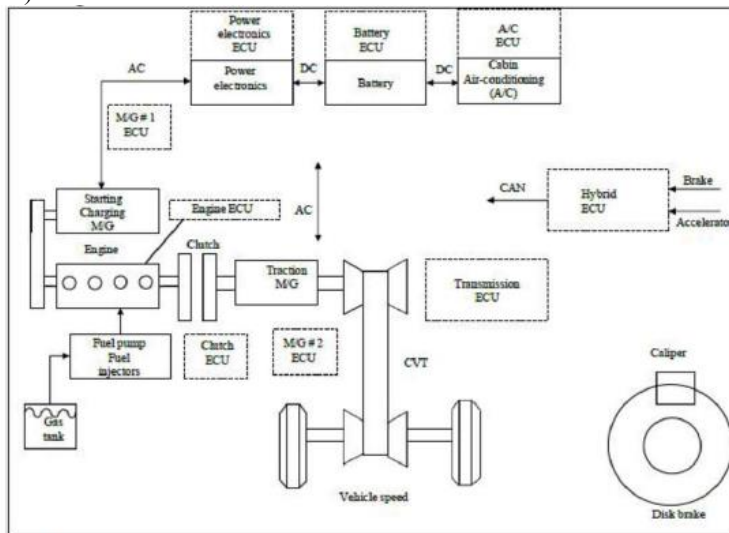
Disadvantages – Two points – 2 marks

b) The main purpose of the Equalizer is to equalize the charge states in your system's batteries. In a 12/24V dual voltage system, the Equalizer output voltage is half of the input voltage. To maintain this ratio, the Equalizer balances the voltage across the batteries

PART C

7. A) Sizing of battery starts with sizing of motors which includes measuring of drive cycles.
Explain
b) Explanation for selection of power electronic circuits
8. A) Batteries, 2. Fuel Cells, 3. Solar Photovoltaic, 4. Flywheels, 5. Supper Capacitors
b) Explain fuzzy logic basics
Fuzzification and Defuzzification

A)



- b) The major functions of the control system are:
to maximize the fuel efficiency – 2 marks (with explanation)
ii. to minimize the exhaust emissions. – 2 marks (with explanation)
b)